retain its execute permissions when added to the image. However, this won't work on Windows because file permissions are handled differently. Instead, we can set this file's permission in our Dockerfile.prod file itself (it's not a bad thing for this to be explicit anyway, whichever platform you're on):

```
COPY . /usr/src/app/
RUN ["chmod", "+x", "/usr/src/app/wait-for"]
```

OK, time to rebuild our image. Remember, though, that our Docker Engine is currently targeting local-vm-1. Since this VM is simulating our production environment, it doesn't make sense to rebuild our image there; we normally build images either locally or as part of our CI pipeline. So let's quickly switch back to targeting our local Docker Engine:

```
$ eval $(docker-machine env -u)
```

Now let's rebuild our image with these changes and push it to DockerHub:

```
$ docker build -f Dockerfile.prod -t robisenberg/myapp_web:prod .
$ docker push robisenberg/myapp_web:prod
```

Now we're ready to define our one-shot db-migrator service in our docker-stack.yml file:

```
Line 1  db-migrator:
  2       image: robisenberg/myapp_web:prod
  3       command: ["./wait-for", "--timeout=300", "database:5432", "--",
  4                 "bin/rails", "db:migrate"]
  5       env_file:
  6         - .env/production/database
  7         - .env/production/web
  8       deploy:
  9         restart_policy:
 10           condition: none
```

The command (lines 3–4) says, "Wait for up to 5 mins (300 seconds) until the host called database is accepting connections on port 5432, and as soon as it's available, run the command bin/rails db:migrate." This means that our database will be migrated successfully even if it takes a considerable time for the database to become available.

The restart_policy (line 9) prevents Swarm from trying to restart it once it has terminated; this setting is what gives the service its one-shot nature.

Sorted. Now our app can be deployed with the database fully migrated.

Deploying Our App on a Swarm

With our newly created docker-stack.yml, we have a description of our app that Swarm can use. We're ready to deploy our app onto our (single-node) swarm.